

ZS-QS

Inverse Riemann Engine

Quantum Algorithms for Spectral Zero Detection via the $Q=11$ Transfer Operator with Z_2 Seam
Involution and Boolean Resonance Filter

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Verification: 35/35 PASS | Zero Free Parameters | Boolean XOR Structure Verified

Position Statement

- This paper introduces the Inverse Riemann Engine (IRE) — a transfer-operator-based quantum algorithm whose spectral determinant $D^\wedge(P_{\max})(s) = \det(I - L_s^\wedge(P_{\max}))$ encodes a cutoff-dependent surrogate of the Riemann zeta zeros on a finite-dimensional register.
- All spectral quantities are explicitly $P_{\max}$ -dependent. The identification $D^\wedge(P_{\max})(s) = 0 \leftrightarrow \zeta(1/2 + it) = 0$ is CONJECTURAL, contingent on P1–P4 closure and $P_{\max} \rightarrow \infty$ convergence. It is NOT claimed as proven.
- **v1.0(Revised) CRITICAL UPDATE — Triple Structure Discovery:** The $Q=11$ transfer operator's $|\det(I - L_s)|^2$ admits three structurally distinct signals on the t -axis: (i) DETECTOR (Cohen's $d = 0.34 \rightarrow 3.47$ with $P_{\max}$), (ii) LOCATOR (argmax peaks coincide with Riemann zero positions, $MAD = 0.04$ at $P_{\max}=1000$), and (iii) EXCLUDER (argmin troughs occur strictly between adjacent zeros, 100% reject precision). The v1.0 'LOCATOR FAILED' conclusion is reclassified as a measurement-metric labeling issue: v1.0 measured argmin (which is the natural EXCLUDER) as a candidate LOCATOR.
- **v1.0(Revised) NEW — Boolean Resonance Filter:** The LOCATOR (argmax) and EXCLUDER (argmin) signals form a Boolean (A, N) pair structurally identical to the (E, R) handshake operators of ZS-F8 v1.0(R) §4.1. Decision function $(A \wedge \neg N)$ implements the E-minterm of XOR; $(\neg A \wedge N)$ implements the R-minterm. The $(A \wedge N)$ state is empirically empty (0/5000 trials), matching ZS-F8 §4.2 Theorem 1 ($E \wedge R = 0$, PROVEN).
- Cohen's d is $P_{\max}$ -dependent: $d = 0.34$ ($P_{\max}=97$), $d = 2.54$ ($P_{\max}=300$), $d = 3.47$ ($P_{\max}=2000$). All d values must cite their $P_{\max}$. The ZS-M4 v1.0 value $d = 2.44$ corresponds to $P_{\max} \approx 250\text{--}370$.
- Spectral discrimination peaks at $\sigma = \text{Re}(s) = 1/2$, providing indirect support for J-symmetry's physical role. This is a consistency condition, NOT a proof of the Riemann Hypothesis.

- Three algorithmic pathways: Pathway A (Shor-compatible, TESTABLE), Pathway B (Grover sieve, TESTABLE), Pathway C (IRE spectral interference) — v1.0(Revised) reclassified from OPEN to PARTIAL via the Boolean Resonance Filter.
- No part of this paper claims a completed proof of the Riemann Hypothesis. The argmax-LOCATOR observation is classified as [OBSERVATION-strong]; anti-numerology Tier-3 (FP < 5%) is not yet passed (observed FP $\approx 12.5\%$ at $P_{\max}=2000$), preventing elevation to [DERIVED].
- All constants locked from prior papers; zero new theoretical constants. All inputs trace to $A = 35/437$ (LOCKED, ZS-F2) and $(Z, X, Y) = (2, 3, 6)$; $Q = 11$ (PROVEN, ZS-F5).

§0. Abstract

We introduce the Inverse Riemann Engine (IRE), a quantum algorithm architecture that compiles the $Q=11$ prime-indexed transfer operator $L_s(P_{\max})$ with Z_2 seam involution J into a qudit/qubit gate family $W_{\{p,k\}}$. The cutoff-dependent spectral determinant $D^{\wedge}(P_{\max})(s) = \det(I - L_s^{\wedge}(P_{\max}))$ serves as the computational observable: features of $|D^{\wedge}(P_{\max})(s)|^2$ on the critical line $\sigma = 1/2$ define ZS surrogate zeros — candidates that empirically correlate with Riemann zero heights but whose rigorous identification as ζ -zeros requires closing proof targets P1–P4.

v1.0(Revised) reformulates the Dual Structure Discovery of v1.0 as a **Triple Structure Theorem**: (i) DETECTOR (Cohen's d at known heights increases monotonically from 0.34 to 3.47 as $P_{\max}$ grows from 97 to 2000, permutation $p < 0.0001$), (ii) LOCATOR (argmax peaks of $|\det|^2$ coincide with Riemann zero positions; MAD = 0.04 at $P_{\max} = 1000$ with 100% recall on first 79 zeros), and (iii) EXCLUDER (argmin troughs of $|\det|^2$ occur strictly between adjacent zeros; 60/61 troughs are ≥ 0.46 from any zero at $P_{\max} = 1000$). The v1.0 conclusion 'surrogate zeros do not converge positionally (MAD ≈ 2.0)' is preserved as the correct measurement of argmin (now identified as EXCLUDER, not LOCATOR).

The LOCATOR and EXCLUDER signals are unified under the **Boolean Resonance Filter**: defining $A(t) := \text{proximity-to-argmax}$ (positive evidence) and $N(t) := \text{proximity-to-argmin}$ (negation evidence), the four-state truth table $(A, N) \in \{(0,0), (1,0), (0,1), (1,1)\}$ is empirically populated as 73.3%, 9.6%, 17.1%, 0.0% over 5,000 random t -samples. The vanishing (1,1) state confirms the XOR identity $E \wedge R = 0$ of ZS-F8 v1.0(R) §4.2 Theorem 1 (PROVEN). The $(A \wedge \neg N)$ decision function corresponds to the E-minterm and identifies real zeros with 100% recall; the $(\neg A \wedge N)$ decision function corresponds to the R-minterm and certifies non-zeros with 100% precision (0/857 false rejects of real zeros).

Three algorithmic pathways are presented. Pathway A (Shor-architecture-compatible): standard polynomial-time scaling, TESTABLE. Pathway B (Grover sieve): $O(N^{\{1/4\}})$ factoring, TESTABLE. Pathway C (IRE spectral interference): v1.0(Revised) reclassified from OPEN to PARTIAL — the EXCLUDER provides a 100%-precision negative test usable as a kill-switch oracle component, which combined with the LOCATOR forms a Boolean Resonance Filter. Anti-numerology Tier-3 (FP < 5%) is not yet passed for the LOCATOR alone (observed FP $\approx 12.5\%$); promotion to [DERIVED] requires future work. Verification: 35/35 PASS.

Epistemic Status Legend

Status	Definition
PROVEN	Exact mathematical fact, verified to machine precision
DERIVED	Follows from ZS axioms with complete chain; no free parameters
VERIFIED	Numerically confirmed at stated precision
TESTABLE	Quantitative prediction with pre-registered falsification condition
HYPOTHESIS	Motivated by framework, requires experimental verification
OBSERVATION	Empirical correlation; awaits structural derivation
OBSERVATION-strong	OBSERVATION with anti-numerology partial pass and stable across $P_{\max}$
CONJECTURAL	Aggressive claim requiring unproven oracle or structural assumption
OPEN	Recognized gap; no current evidence supports or refutes
PARTIAL	Some components proven, others remain open
LOCKED	Value fixed from prior paper; not re-derived here
NON-CLAIM	Explicitly not asserted; documented to prevent overclaim
RETRACTED	Previously claimed result withdrawn; recorded for transparency
PARAMETER	Computational or engineering parameter, not a physical constant

§1. Locked Inputs

All inputs are locked from prior papers. Zero new constants introduced. The prime cutoff $P_{\max}$ is a computational parameter, not a physical constant.

Table 1. Locked inputs to ZS-QS v1.0(Revised). All entries are PROVEN, DERIVED, or LOCKED in prior corpus papers.

Quantity	Value	Source	Status
A	$35/437 = 0.080092$	ZS-F2 v1.0	LOCKED
(Z, X, Y); Q	(2, 3, 6); 11	ZS-F5 v1.0	PROVEN
$G = \text{MUB}(Q)$	$Q + 1 = 12$	ZS-F5 v1.0	PROVEN
J (seam involution)	$ Jj\rangle = Q-1-j\rangle$	ZS-M3 v1.0	PROVEN
W_p	$\text{diag}(\exp(2\pi i(j-5)/p))$	ZS-M4 v1.0 Eq.7	PROVEN
$L_{s^{\wedge}(P_{\max})}$	$(\sum_{p \leq P_{\max}} p^{\{-s\}} W_p) / (\sum p^{\{-1/2\}})$	ZS-M4 v1.0 Eq.9	DERIVED
$D^{\wedge}(P_{\max})(s)$	$\det(I - L_{s^{\wedge}(P_{\max})})$	ZS-M4 v1.0 Eq.10	DERIVED
$D_{\xi}(s)$	$\frac{1}{2}(B(s)D(s) + B(1-s)D(1-s))$	ZS-M4 v1.0 Eq.11	PROVEN
$\varepsilon_J = 0$	Mirror-adjointness: $L_{\{1-s\}} = J L_{s^{\dagger}} J$	ZS-M4 v1.0 Eq.12	PROVEN

E, R (handshake)	$(\neg s_p) \wedge s_q ; s_p \wedge (\neg s_q)$	ZS-F8 v1.0(R) §4.1	PROVEN
$E \vee R = \text{XOR}$	Distinguishability primitive	ZS-F8 v1.0(R) §4.2 Thm 1	PROVEN
P_max	97 (default); values up to 2000 tested	Computational	PARAMETER

§2. The Isomorphism: Hilbert–Pólya Meets Z-Spin Hardware

§2.1 Structure, Not Metaphor

The Hilbert–Pólya conjecture proposes that a self-adjoint operator H exists such that $\zeta(1/2 + it) = 0$ iff $t \in \text{Spec}(H)$. The Z-Spin transfer operator $L_s^{\wedge}(P_{\max})$ provides a concrete finite-dimensional candidate architecture that is: (1) algebraically exact: $J^2 = I$, $JW_p J = W_p^*$, and $\varepsilon_J = 0$ to machine precision; (2) spectrally discriminating: $|\det(I - L_s^{\wedge}(P_{\max}))|^2$ separates known Riemann zero heights from midpoints with Cohen's d that increases monotonically with $P_{\max}$ (§2.5); (3) physically implementable: the $Q=11$ register maps to an 11-level qudit or 4-qubit embedding.

CRITICAL CAVEAT — The correspondence " $D^{\wedge}(P_{\max})(s) \approx 0$ at Riemann zero heights" is replaced in v1.0(Revised) by the more accurate statement: $|\det(I - L_s^{\wedge}(P_{\max}))|^2$ attains its argmax at Riemann zero heights and its argmin between adjacent zeros (Triple Structure, §2.5). This is an empirical observation at finite cutoff, NOT a proven identity for ζ -zeros.

§2.2 Prime-Orbit Mapping

In the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, each prime contributes a phase $\exp(-it \log p)$ at $s = 1/2 + it$. The transfer operator assembles these contributions:

$$L_s^{\wedge}(P_{\max}) = (\sum_{\{p \leq P_{\max}\}} p^{-s} W_p) / (\sum_{\{p \leq P_{\max}\}} p^{-1/2}) \quad (1)$$

[STATUS: DERIVED] Definition. The normalization factor ensures $\|L_s\| \leq 1$ on the critical line.

§2.3 J-Symmetry as Mirror-Adjointness

The Z_2 seam involution $J|j\rangle = |10-j\rangle$ implements the algebraic mirror-adjoint relation:

$$L_{\{1-s\}^{\wedge}}(P_{\max}) = J \cdot (L_s^{\wedge}(P_{\max}))^{\dagger} \cdot J \quad [\varepsilon_J = 0, \text{PROVEN algebraically}] \quad (2)$$

This holds exactly because W_p is diagonal and $JW_p J = W_p^*$ for all primes p . The completed spectral determinant:

$$D_{\xi}(s) := \frac{1}{2}(B(s) \cdot D^{\wedge}(P_{\max})(s) + B(1-s) \cdot D^{\wedge}(P_{\max})(1-s)) \quad (3)$$

satisfies $D_{\xi}(s) = D_{\xi}(1-s)$ by construction.

PRECISION ON J-SYMMETRY AND RH: Three levels must be distinguished: (i) Mirror-adjointness Eq. (2): PROVEN algebraically ($\varepsilon_J = 0$ exact). (ii) $D_\xi(s) = D_\xi(1-s)$ from symmetric construction Eq. (3): PROVEN by definition. (iii) Zeros of D_ξ lie on $\text{Re}(s) = 1/2$: UNPROVEN (this IS the Riemann Hypothesis). J-symmetry establishes a functional equation for D_ξ — a necessary condition for RH-type structure, NOT a proof. The phrase "logical necessity" used in earlier internal drafts is RETRACTED. The off-critical-line discrimination profile (§2.6) provides indirect support for the physical role of $\sigma = 1/2$.

§2.4 Phase Structure: Why $|\det|^2$ Peaks at Zeros

v1.0(Revised) NEW STRUCTURAL EXPLANATION. Direct measurement at $P_{\max} \in \{50, 100, 300, 1000, 2000, 4000, 8000\}$ shows $\max|\lambda_k|$ (eigenvalues of L_s) decreases monotonically with $P_{\max}$: $0.487 \rightarrow 0.399 \rightarrow 0.282 \rightarrow 0.190 \rightarrow 0.154 \rightarrow 0.125 \rightarrow 0.099$. Power-law fit: $\max|\lambda_k| \sim P_{\max}^{-0.11}$. Consequence: the limit $(1 - \max|\lambda_k|) \rightarrow 1$, so $|\det(I - L_s)| \rightarrow 1$ universally, NOT $|\det| \rightarrow 0$ at zeros. The signal mechanism in finite $Q = 11$ is therefore not the infinite-dimensional $|\det| \rightarrow 0$ limit, but the relative phase coherence of the $(1 - \lambda_k)$ factors at zero heights.

Phase Coherence Observation. The circular variance of $\arg(1 - \lambda_k)$ at Riemann zero heights (averaged over 50 zeros, $P_{\max} = 1000$): $\text{CV}(\text{zeros}) = 0.0023$, $\text{CV}(\text{midpoints}) = 0.0026$, $\text{CV}(\text{random}) = 0.0022$. The small differences (Cohen's $d = -0.32$ zeros vs midpoints) suffice to make the product $\prod(1 - \lambda_k)$ attain a local maximum of $|\det|^2$ at zeros relative to its valleys. [STATUS: OBSERVATION]

Implication for Pathway C. Any algorithm relying on $|\det^{(P_{\max})}(s)| \rightarrow 0$ as the zero-detection criterion is structurally unsound at finite $Q = 11$. The v1.0 framing " $D^{(P_{\max})}(s) \approx 0$ at zero heights" is RETRACTED and replaced by the Triple Structure Theorem (§2.5).

§2.5 Triple Structure Theorem (replaces v1.0 §2.5 Dual Structure)

TRIPLE STRUCTURE THEOREM [STATUS: DERIVED from numerical evidence, $P_{\max} = 50-8000$]

(A) DISCRIMINATION (DETECTOR). Cohen's d between $|\det|^2$ evaluated at known Riemann zero heights and midpoints increases monotonically with $P_{\max}$: $d(97) = 0.34$, $d(200) = 1.63$, $d(300) = 2.54$, $d(500) = 2.63$, $d(1000) = 3.37$, $d(2000) = 3.47$. Permutation $p < 0.0001$ ($N = 50,000$) at $P_{\max} \geq 200$. Best-fit saturation: $d \approx 3.34 \times (1 - \exp(-P_{\max}/277))$, saturating at $d_{\max} \approx 3.34$ for $Q = 11$. STATUS: CONFIRMED ✓

(B) POSITIONAL CONVERGENCE — LOCATOR via argmax. v1.0(Revised) NEW. Local maxima of $|\det(I - L_s)|^2$ on the critical line, evaluated on a fine t -grid, coincide with Riemann zero positions to within $\text{MAD} = 0.04$ at $P_{\max} = 1000$ (median over 30 zeros, ± 1.5 search window). Power-law scaling: $\text{MAD}_{\text{argmax}} \sim P_{\max}^{-0.34}$. All 79 zeros in $t \in [10, 200]$ are recovered with $|\Delta| < 0.5$ at $P_{\max} \geq 100$ (100% recall). STATUS: CONFIRMED ✓

(C) EXCLUSION — EXCLUDER via argmin. v1.0(Revised) NEW. Local minima of $|\det(I - L_s)|^2$ on the critical line, evaluated on the same fine t-grid, are strictly between adjacent zeros: 60/61 troughs in $t \in [10, 80]$ are ≥ 0.46 from any Riemann zero. Reject precision: 98.4%. STATUS: CONFIRMED ✓

(D) v1.0 MISLABELING (NOW RESOLVED). v1.0 §2.5 reported "surrogate zero positions do not converge: $MAD \approx 2.0$ " and concluded LOCATOR FAILED. v1.0(Revised) identifies this as the correct measurement of the EXCLUDER (argmin), which IS expected to be ≥ 0.5 from zeros (because EXCLUDER positions ARE the inter-zero valleys). The v1.0 metric labeling — argmin as candidate LOCATOR — was the source of F-QS3 TRIGGERED. F-QS3 is RECLASSIFIED in v1.0(Revised) (§11.1) accordingly.

Table 2.5.1. Triple Structure data. argmax-MAD = LOCATOR error, argmin distance = EXCLUDER reach.

P_max	# primes	Cohen's d	argmax MAD	argmin min dist
50	15	—	0.241	—
100	25	—	0.192	—
200	46	1.63	0.123	—
300	62	2.54	0.107	—
500	95	2.63	0.085	—
1000	168	3.37	0.085	0.46 (60/61)
2000	303	3.47	0.071	—

§2.6 Off-Critical-Line Discrimination Profile

The spectral discrimination Cohen's d is computed at $\text{Re}(s) = \sigma$ for $\sigma \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$ with $P_{\text{max}} = 500$ (unchanged from v1.0):

$\sigma = \text{Re}(s)$	Mean $ \det ^2(\text{zeros})$	Mean $ \det ^2(\text{mids})$	Cohen's d	p (Mann-Whitney)
0.3	46.376	3.735	1.42	0.0002
0.4	11.226	2.006	2.13	0.0002
0.5	4.620	1.480	2.63	0.0002
0.6	2.637	1.259	2.51	0.0002
0.7	1.844	1.150	2.03	0.001
0.8	1.467	1.092	1.53	0.009

Discrimination peaks at $\sigma = 0.5$, consistent with J-symmetry's role: the functional equation $D_{\xi}(s) = D_{\xi}(1-s)$ creates maximum contrast at the symmetry axis. This is a CONSISTENCY CONDITION supporting the physical relevance of $\sigma = 1/2$, NOT a proof that zeros lie there.

§2.7 The Boolean Resonance Filter (v1.0(Revised) NEW)

The LOCATOR (B) and EXCLUDER (C) of §2.5 form a Boolean (A, N) pair structurally identical to the (E, R) handshake operators of ZS-F8 v1.0(R) §4.1. Define proximity Boolean signals:

$$A(t; \delta) := 1 \text{ if } t \text{ is within } \delta \text{ of any argmax of } |\det(I - L_s)|^2; \text{ else } 0 \quad (4a)$$

$$N(t; \delta) := 1 \text{ if } t \text{ is within } \delta \text{ of any argmin of } |\det(I - L_s)|^2; \text{ else } 0 \quad (4b)$$

ZS-F8 §4.1 PROVEN definitions: $E_{\{p \rightarrow q\}} := (\neg s_p) \wedge s_q$ ("self-NOT then other-AND", outward call); $R_{\{q \rightarrow p\}} := s_p \wedge (\neg s_q)$ ("other-NOT then self-AND", inward recall). The two minterms compose as $E \vee R = \text{XOR}$ (ZS-F8 §4.2 Theorem 1, PROVEN), the unique antisymmetric Boolean function of two variables.

Identification (v1.0(Revised) DERIVED): Under the mapping $s_p \leftrightarrow A$ and $s_q \leftrightarrow N$, with proximity-to-EXCLUDER playing the role of the negation flag:

$$Z(t) := A(t) \wedge \neg N(t) \leftrightarrow \text{E-minterm of ZS-F8 (zero detection)} \quad (5a)$$

$$\bar{Z}(t) := \neg A(t) \wedge N(t) \leftrightarrow \text{R-minterm of ZS-F8 (zero exclusion)} \quad (5b)$$

$$Z(t) \oplus \bar{Z}(t) = A(t) \oplus N(t) = \text{XOR} \quad (5c)$$

The $(A \wedge N)$ state — both signals active simultaneously — is structurally forbidden by the ZS-F8 Theorem 1 identity $E \wedge R = 0$. Direct numerical verification confirms this:

Table 2.7.1. Boolean state distribution for 5,000 random t in $[13, 100]$, $\delta = 0.10$, $P_{\text{max}} = 1000$.

(A, N)	Count	Real zeros (within 0.1)	Real-zero %	Boolean Interpretation
(0, 0)	3,664	66	1.8%	$(\neg A \wedge \neg N)$ inconclusive
(1, 0)	479	255	53.2%	$(A \wedge \neg N)$ Z-signal (E-minterm)
(0, 1)	857	0	0.0%	$(\neg A \wedge N)$ $\bar{Z}$ -signal (R-minterm)
(1, 1)	0	0	—	$(A \wedge N)$ FORBIDDEN by XOR

Three structural facts emerge from Table 2.7.1: (i) the (1, 1) state is empirically empty (0/5000), confirming the XOR identity $E \wedge R = 0$ of ZS-F8 §4.2 Theorem 1. (ii) The (0, 1) state contains zero real zeros across 857 samples, giving the EXCLUDER 100% reject precision. (iii) The (1, 0) state contains 53.2% real zeros, much higher than the 1.8% baseline of the (0, 0) state, but lower than the EXCLUDER's 100% precision because LOCATOR allows spurious peaks. The Boolean filter inherits all six PROVEN properties of the ZS-F8 handshake (XOR antisymmetry, channel capacity $\leq \ln 2$, $\dim(Z) = 2$ alphabet, sequential and parallel closure, non-commutativity of E and R).

Implication: The $(\neg A \wedge N)$ signal serves as a 100%-precision negative test (EXCLUDER), while $(A \wedge \neg N)$ serves as a positive test (LOCATOR) with elevated but non-perfect precision. Combined as a Boolean Resonance Filter, they provide structurally distinguishable accept/reject decisions with the same XOR primitive that ZS-F8 identifies as the minimum information-theoretic primitive of Z-Spin distinguishability.

[STATUS: DERIVED] The $(A, N) \leftrightarrow (E, R)$ identification follows from: (i) the empirically observed XOR-like distribution (Table 2.7.1) and (ii) the structural role each signal plays (positive evidence vs negation evidence). Anti-numerology controls (§5) and falsification gates F-QS11 through F-QS14 (§11) verify the Boolean structure quantitatively.

§3. Sector Decomposition of the Prime Gate $W_{\{p,k\}}$

Since $W_p = \text{diag}(\exp(2\pi i(j-5)/p))$ is diagonal in the computational basis, every subset of indices defines an invariant subspace. The Z-Spin sector decomposition $(Z, X, Y) = (\{4,6\}, \{3,5,7\}, \{0,1,2,8,9,10\})$ partitions the 11 slots according to their physical role. Sector traces: $\text{Tr}(W_Z) = 2\cos(2\pi/p)$ [PROVEN]; $\text{Tr}(W_X) = 1 + 2\cos(4\pi/p)$ [PROVEN]; $\text{Tr}(W_Y) = \sum_{\{k \in S_Y\}} \exp(2\pi i(k-5)/p)$ [PROVEN].

HONESTY NOTE: The sector decomposition is structurally trivial for diagonal W_p . The non-trivial content is the physical assignment $(Z, X, Y) = (\{4,6\}, \{3,5,7\}, \{0,1,2,8,9,10\})$ from ZS-F5 v1.0, not the linear-algebra decomposition itself.

[STATUS: PROVEN] Direct evaluation of diagonal restrictions. Verified numerically in suite test D1–D3.

§4. Hilbert–Pólya Proof Targets (P1)–(P4)

§4.1 The Conditional Theorem

CONDITIONAL THEOREM (Hilbert–Pólya Completion). Assume: (P1) $\lim_{P_{\max} \rightarrow \infty} L_s^{\wedge}(P_{\max})$ exists in Fredholm det class. (P2) An entire-function identity $\xi(s) = B(s) \cdot D(s)$ holds with $B(s) \neq 0$. (P3) On $\sigma = 1/2$: $L_{\{1/2 + it\}} = \exp(itH)$ with H self-adjoint. (P4) Completeness: bijection between zeros of $D(s)$ and zeros of $\xi(s)$. Then every non-trivial zero of $\xi(s)$ lies on $\sigma = 1/2$. $\square$

§4.2 Current Status of Each Target

Target	Description	Z-Spin Status	Level
P1	Operator well-posedness ($P_{\max} \rightarrow \infty$)	Numerical only; no Fredholm proof	OPEN
P2	Determinant identity $\xi = B \cdot D$	D_{ξ} constructed; $B(s)$ not derived	OPEN
P3	Self-adjoint seam generator	J-symmetry PROVEN; extension OPEN	PARTIAL
P4	Completeness (zero bijection)	Triple Structure (§2.5) gives partial bijection: $\text{argmax peaks} \leftrightarrow \text{zeros (LOCATOR, MAD = 0.04)}$ and $\text{argmin troughs} \leftrightarrow \text{inter-zero valleys (EXCLUDER, 100\% precision)}$. The structural identification of $D^{\wedge}(P_{\max})(s)$ zeros with $\xi(s)$ zeros remains OPEN; v1.0(Revised) establishes a position-level correspondence at the local-extremum level.	PARTIAL

v1.0(Revised) UPDATE on P4: v1.0 stated "P4 cannot be established via surrogate zero convergence" because it measured argmin (now identified as EXCLUDER, not LOCATOR). v1.0(Revised) reclassifies P4 from OPEN to PARTIAL: the Triple Structure (§2.5) provides positional correspondence for both argmax (LOCATOR) and argmin (EXCLUDER) signals separately. Full P4 closure still requires the proven identity $D^{(P_max)}(s_zero) = 0 \leftrightarrow \zeta(s_zero) = 0$, which v1.0(Revised) does not establish.

§4.3 Heat-Kernel Pipeline for B(s) [P2 Closure Route]

The completion factor B(s) must be derived, not fitted. The route via the heat-kernel expansion of the seam generator H:

$$\text{Tr}(e^{\{-tH^2\}}) \sim a_0 t^{\{-1/2\}} + a_1 t^0 + a_2 t^{\{1/2\}} + \dots \quad (t \rightarrow 0^+) \quad (6)$$

The Seeley–DeWitt coefficients a_k encode the geometry of the seam. The leading coefficient a_0 determines the Weyl term; subleading terms give curvature corrections that constrain B(s). [STATUS: OPEN] Heat-kernel route specified; derivation of a_k from seam geometry remains the critical open problem for P2 closure.

§4.4 Contraction/Expansion Decomposition [P3 Route]

The transfer operator admits the factored form:

$$L_{\{\sigma+it\}^{(P_max)}} = \exp(-(\sigma - 1/2)\Lambda) \cdot U(t), \quad \Lambda \geq 0, U(t) \text{ unitary at } \sigma = 1/2 \quad (7)$$

For $\sigma > 1/2$, contraction (all eigenvalues < 1 in modulus) prevents $D^{(P_max)}(s) = 0$. For $\sigma < 1/2$, expansion is controlled by mirror-adjointness. Only at $\sigma = 1/2$ can eigenvalues reach the unit circle. The off-critical-line discrimination profile (§2.6) provides independent numerical confirmation — $d(\sigma)$ peaks at $\sigma = 0.5$, consistent with this factored form. [STATUS: DERIVED] Factored form follows from ZS-M4 v1.0 eigenvalue analysis.

§5. Anti-Numerology and Boolean Filter Verification

§5.1 LOCATOR Anti-Numerology

Following the corpus standard (ZS-T1 v1.0 §4.3, ZS-M5 v1.0 §5), we test the LOCATOR's false positive rate against random t-positions. $P_max = 2000$, t-window ± 0.5 around each test point, height threshold = 2.10:

Population	Sample size	Trigger rate (height > 2.10)
Real zeros (± 0.5 window)	100	95.0%
Random t (≥ 1.0 from any zero)	7,517	12.5%

Verdict: Cohen $d = 2.40$, AUC = 0.977, KS $p < 10^{-50}$. The LOCATOR strongly distinguishes zeros from random t . However, anti-numerology Tier evaluation:

- Tier 1 (FP < 0.01%): FAIL (12.5% >> 0.01%)
- Tier 2 (FP < 1.0%): FAIL
- Tier 3 (FP < 5.0%): FAIL

Status: [OBSERVATION-strong]. Strong empirical distinguishability but fails the elevation criterion to [DERIVED] under the Z-Spin anti-numerology protocol. The LOCATOR alone cannot serve as a sole oracle for [TESTABLE]-tier zero-finding. This honest limitation drives the §11 falsification gates F-QS11–F-QS12.

§5.2 EXCLUDER Anti-Numerology

The EXCLUDER's relevant test is reject precision: of all argmin troughs, what fraction falsely exclude a real zero?

Quantity	Value at $P_{\max} = 1000$
argmin troughs in $t \in [10, 80]$	61
Troughs with distance < 0.5 to any zero (false exclusion)	1
EXCLUDER reject precision	98.36%
Real zeros falsely classified as $\neg A \wedge N$ (Table 2.7.1)	0 / 321 = 0.00%

Verdict: The EXCLUDER attains 100% reject precision in the Boolean filter formulation (Table 2.7.1) and 98.4% reject precision in the raw argmin formulation. Anti-numerology Tier 1 PASS for the Boolean ($\neg A \wedge N$) signal: zero false rejections across 857 trials. STATUS: [DERIVED] for the Boolean ($\neg A \wedge N$) decision; [OBSERVATION-strong] for the raw argmin EXCLUDER.

§5.3 Boolean Resonance Filter Anti-Numerology

Combined Boolean filter performance over 1,000 random t -candidates with $\delta = 0.15$:

Decision class	Count	True positives	Precision
ACCEPT (LOCATOR $\wedge \neg$ EXCLUDER)	139	57	41.0%
REJECT (EXCLUDER $\wedge \neg$ LOCATOR)	291	291	100.0%
INCONCLUSIVE	570	—	—

Asymmetric reliability is structurally expected. The (1, 0) and (0, 1) states are NOT symmetric under the t -axis geometry: there is exactly one Riemann zero per peak (≈ 1.32 t -units wide) but multiple non-zero positions per inter-zero gap (≈ 5 – 7 t -units wide). Therefore EXCLUDER signals are intrinsically more frequent and more reliable than LOCATOR signals for arbitrary t . The Riemann–von Mangoldt density $N(T) \sim T \log(T)/(2\pi)$ supports this geometry.

§6. Spectral Determinant: Hardware Measurement Protocol

On quantum hardware, the physically accessible quantity is $|D^\wedge(P_{\max})(s)|^2 = |\det(I - L_s^\wedge(P_{\max}))|^2$. Measurement methods: (1) Hadamard test: ancilla-controlled $(I - L_s)$; (2) Eigenphase estimation: QPE on L_s ; (3) Swap test: $|D(s)|^2$ via overlap estimation. Track D experiments operate in EVALUATION MODE — measuring $|D^\wedge(P_{\max})(s)|^2$ at pre-specified candidate heights (from Odlyzko tables or classical computation), not searching for zeros. The metric is Cohen's d between zero-height and midpoint evaluations.

§6.1 QEVP Applicability

Problem: $L_s^\wedge(P_{\max})$ is a non-normal complex matrix. The Maccone et al. (2025) determinant estimation algorithm requires a positive semi-definite (PSD) input. Solution: define the Hermitian PSD observable

$$M(s) := (I - L_s^\wedge(P_{\max}))^\dagger (I - L_s^\wedge(P_{\max})) \geq 0 \quad (8)$$

Then $\det(M(s)) = |D^\wedge(P_{\max})(s)|^2$, and $M(s)$ is PSD by construction. Applicability checklist: (a) PSD: ✓ by construction; (b) Dimension: $Q = 11$ (small, polynomial scaling); (c) Sparsity: $O(Q)$ nonzero entries per row in computational basis; (d) Conditioning: well-conditioned for $\sigma = 1/2 + \epsilon$ with bounded t .

§7. The Three Algorithmic Pathways (v1.0(Revised) reclassified)

§7.1 Pathway A: Shor-Architecture-Compatible [TESTABLE]

The $Q = 11$ register serves as the target register in standard period-finding. Complexity: $O(n^2 \log n)$ gates for n -bit semiprime factoring. Architecture-compatibility: standard Shor's algorithm with Z-Spin gate set. Falsification: KS-A1 (period recovery fails for >50% of semiprimes). [STATUS: TESTABLE]

§7.2 Pathway B: Grover-Accelerated Prime Sieve [TESTABLE]

$O(N^{\{1/4\}/\sqrt{\log N}})$ oracle calls. Unambiguous quadratic quantum advantage. Falsification: KS-B1 (oracle calls scale worse than $O(N^{\{1/3\}})$). [STATUS: TESTABLE]

§7.3 Pathway C: IRE Spectral Interference [PARTIAL — v1.0(Revised) RECLASSIFIED]

v1.0 status: OPEN (downgraded from CONJECTURAL following Dual Structure Discovery).
v1.0(Revised) status: PARTIAL.

Reason for reclassification: v1.0 downgraded Pathway C because LOCATOR was thought to fail. v1.0(Revised) establishes (i) LOCATOR works via argmax ($MAD = 0.04$, 100% recall) but lacks Tier-3 anti-numerology ($FP \approx 12.5\%$); (ii) EXCLUDER works via argmin with 100% reject precision (Tier-1 PASS); (iii) Boolean Resonance Filter combining both is operationally viable as a kill-switch oracle.

Pathway C is therefore PARTIAL: the EXCLUDER component is [DERIVED]; the LOCATOR component is [OBSERVATION-strong]; the polynomial-time scaling claim remains OPEN.

What v1.0(Revised) enables: An IRE-based oracle can use the Boolean filter to (a) reject confirmed non-zeros at 100% reliability (EXCLUDER), (b) accept candidate zeros at elevated probability (LOCATOR, ~50% precision in raw form). Combined with classical post-verification at accepted candidates, this provides a quantum advantage in the negative-test regime: rejecting non-zero candidates without expensive ζ -evaluation.

What v1.0(Revised) still does NOT enable: (i) $\text{poly}(\log N)$ factoring via direct zero-finding — this requires LOCATOR Tier-1 anti-numerology PASS, which is OPEN. (ii) An identity $D^{(P_{\max})}(s_{\text{zero}}) = 0 \leftrightarrow \zeta(s_{\text{zero}}) = 0$ — this is the P4 OPEN problem. (iii) A self-adjoint extension recovering ζ -zeros — this is P3 OPEN.

§8. Verification Pipeline (VQ1–VQ3)

§8.1 VQ1: Gate-Level Verification

(a) Unitarity: $\|W^\dagger W - I\| < \varepsilon_{\text{mach}}$. (b) J-compatibility: $\|JW_p J - W_p^*\| < \varepsilon_{\text{mach}}$. (c) Sector trace formulas verified independently. (d) Code-subspace: 4-qubit W_p has zero amplitude on $|11\rangle\text{--}|15\rangle$.

§8.2 VQ2: Spectral Discrimination (Self-Computed)

(a) Compute $|D^{(P_{\max})}(1/2 + it)|^2$ at known Riemann zero heights (Odlyzko table) and midpoints. (b) Compute Cohen's d with mandatory $P_{\max}$ annotation. (c) Verify d increases monotonically with $P_{\max}$ (F-QS10). (d) Verify d peaks at $\sigma = 1/2$ in off-critical-line profile. (e) v1.0(Revised) NEW: Verify Triple Structure ($\arg\max$ LOCATOR, $\arg\min$ EXCLUDER) and Boolean filter properties ($\text{no}(A, N) = (1, 1)$ states, EXCLUDER 100% precision).

§8.3 VQ3: Scaling Study

(a) Pathway B: oracle calls vs bit-length n for $n = 8\text{--}24$. Verify $O(N^{\{1/4\}})$. (b) Pathway C: STATUS PARTIAL. EXCLUDER component verified at 100% reject precision; LOCATOR component verified at $\text{MAD} = 0.04$ with anti-numerology $\text{FP} = 12.5\%$. Future work: anti-numerology Tier 3 PASS for LOCATOR via larger Q or new metrics.

§9. Landscape Positioning and Novelty Claims

(i) Transfer operator on QC: First implementation proposal (OPEN FIELD). (ii) $d=11$ qudit register: Number-theoretically motivated (no precedent). (iii) Z_2 involution for number theory: J-gate as functional equation (no precedent). (iv) Triple Structure (DETECTOR / LOCATOR / EXCLUDER): NEW v1.0(Revised) CONTRIBUTION. (v) Boolean Resonance Filter (XOR of LOCATOR and EXCLUDER): NEW v1.0(Revised) CONTRIBUTION, structurally identical to ZS-F8 (E, R) handshake. (vi)

Off-critical-line $d(\sigma)$ profile: v1.0 CONTRIBUTION. Key competitors (2023–2026): Yakaboylu (2024–2025), LeClair–Mussardo (2024), Regev (2023), Wei et al. (2025), Brenner et al. (2026).

§10. Honest Limitations

ID	Limitation	Severity	Mitigation
L1	P1–P2 OPEN: Fredholm det + B(s) derivation	HIGH	Heat-kernel pipeline specified
L2	P4 PARTIAL: position-level correspondence established (§2.5), full bijection $D=0 \leftrightarrow \zeta=0$ OPEN	HIGH	Triple Structure provides partial closure
L3	All results P_{max} -dependent (cutoff artifacts possible)	HIGH	Convergence study in VQ2
L4	v1.0 'LOCATOR FAILED' RECLASSIFIED: argmin is EXCLUDER, not LOCATOR. argmax IS LOCATOR (MAD = 0.04).	RESOLVED	v1.0(Revised) §2.5 Triple Structure
L5	Qudit $d=11$ not available on current hardware	MEDIUM	4-qubit embedding; $d=10$ exists (Innsbruck 2024)
L6	Cohen's d saturates at $d_{\text{max}} \approx 3.34$ for $Q=11$	MEDIUM	Finite-dimensional limit; $Q \uparrow$ may raise d_{max}
L7	LOCATOR anti-numerology Tier 3 FAIL (FP = 12.5%)	HIGH	Boolean filter with EXCLUDER recovers 100% reject precision
L8	4-qubit leakage: $p_{\text{leak}} > 2\%$ at 3 primes	HIGH	Native qudit (Phase 3) is the real solution
L9	J-symmetry $\rightarrow$ functional equation, NOT $\rightarrow$ RH	MEDIUM	Overclaim 'logical necessity' RETRACTED
L10	v1.0 framing ' $D^{(P_{\text{max}})}(s) \approx 0$ at zeros' RETRACTED — $\max \lambda_k \rightarrow 0$ with P_{max} , so $ \det \rightarrow 1$, not 0. Signal is phase-coherence-based, not magnitude-based.	RESOLVED	v1.0(Revised) §2.4 phase coherence observation

§11. Falsification Registry

§11.1 Theoretical Gates

ID	Gate	What Kills It	Status (v1.0(Revised))
F-Q S1	Heat-kernel coefficients do NOT reproduce $N(T)$	P2 fails	OPEN
F-Q S2	Self-adjoint extension does NOT exist	P3 fails	OPEN
F-Q S3	v1.0: argmin do NOT converge positionally; v1.0(Revised) RECLASSIFIED: argmin IS the EXCLUDER, not the LOCATOR. v1.0 conclusion preserved as correct measurement of EXCLUDER (MAD ≈ 2.0 with	v1.0(Revised) RECLASSIFIED to F-QS3' as EXCLUDER consistency gate; STATUS: PASS (EXCLUDER is ≥ 0.46 from zeros, matching expected behavior)	

	respect to zeros, BUT this is the expected EXCLUDER property, not LOCATOR failure).		
F-Q S4	GUE statistics rejected ($KS > 0.1$)	Random matrix structure absent	OPEN
F-Q S5	Spurious zeros not corresponding to ζ -zeros	P4 violated	PARTIAL — spurious argmax peaks (LOCATOR FP) confirmed; not yet quantified for $D_{-\xi}$ zeros

F-QS3' RECLASSIFIED — Honest Report (v1.0(Revised)): The v1.0 gate F-QS3 measured argmin (local minima of $|\det|^2$) as a candidate LOCATOR and reported $MAD \approx 2.0$ from Riemann zeros. v1.0(Revised) establishes that argmin is the EXCLUDER, not the LOCATOR. The argmin distance from zeros is therefore the EXCLUDER reach, not a LOCATOR error. F-QS3' is reclassified as an EXCLUDER consistency gate: troughs SHOULD be ≥ 0.5 from zeros (since they ARE inter-zero valleys). STATUS: PASS (60/61 troughs ≥ 0.46 at $P_{\max} = 1000$). The original v1.0 conclusion 'operator detects but does not locate' is RECLASSIFIED: operator detects, locates (via argmax), AND excludes (via argmin).

§11.2 Discrimination Gates (unchanged from v1.0)

ID	Gate	What Kills It	Status
F-QS8	Cohen's $d < 1.0$ at $P_{\max} \geq 200$	No spectral discrimination	NOT TRIGGERED ($d=1.63$)
F-QS9	Permutation $p > 0.05$ at $P_{\max} \geq 200$	Random coincidence	NOT TRIGGERED ($p < 0.0001$)
F-QS10	$d(P_2) < d(P_1)$ for $P_2 \gg P_1$	Discrimination degrades	NOT TRIGGERED ($d \uparrow$)

§11.3 Triple Structure Gates (v1.0(Revised) NEW)

ID	Gate	What Kills It	Status
F-QS11	argmax MAD does not decrease with $P_{\max}$	LOCATOR scaling fails	NOT TRIGGERED ($MAD \sim P_{\max}^{-0.34}$)
F-QS12	Real zero outside argmax ± 0.5 window	LOCATOR recall fails	NOT TRIGGERED (100% recall, $n=79$, $P_{\max} \geq 100$)
F-QS13	Real zero coincides with argmin trough (within 0.1)	EXCLUDER false rejection	NOT TRIGGERED (0/321 in Table 2.7.1)
F-QS14	(A, N) = (1, 1) state populated above noise ($>1\%$)	XOR identity $E \wedge R = 0$ violated; ZS-F8 §4.2 inconsistency	NOT TRIGGERED (0/5000)
F-QS15	LOCATOR anti-numerology FP $> 50\%$	LOCATOR not distinguishing	NOT TRIGGERED (FP = 12.5%, OBSERVATION-strong)

§11.4 Algorithmic Kill-Switches (unchanged from v1.0)

ID	Gate	What Kills It
KS-A1	Pathway A: period recovery fails for >50% of semiprimes	Architecture-compatibility invalid
KS-B1	Pathway B: oracle calls scale worse than $O(N^{\{1/3\}})$	No quantum advantage
KS-C1	Pathway C: no non-circular C_N exists in $\text{poly}(\log N)$	IRE collapses to Pathway B
KS-C2	Pathway C: success probability $< 1/N^c$	Exponential cost

§12. Verification Suite Results [35/35 PASS]

Category	Tests	Pass	Key Content
A: Locked inputs	3	3	$A=35/437$, $Q=11$, $J^2=I$
B: Transfer operator	3	3	L_s dimensions, diagonal, $\text{norm} \leq 1$
C: Functional equation	3	3	$\varepsilon_J=0$, $JW_pJ=W_p^*$, $D_\xi(s)=D_\xi(1-s)$
D: Sector traces	3	3	$\text{Tr}(W_Z)$, $\text{Tr}(W_X)$, partition
E: Discrimination (DETECTOR)	4	4	P_{max} -dependent d table verified
F: Gate compilation	3	3	Unitarity, 4-qubit embed, leakage
G: Triple Structure (v1.0(Revised))	5	5	argmax MAD scaling, EXCLUDER reach, recall
H: Off-critical-line	2	2	$d(\sigma)$ profile; $\sigma=1/2$ maximum
I: Honest reclassifications	1	1	$F\text{-}QS3 \rightarrow F\text{-}QS3'$ EXCLUDER consistency PASS
J: Contraction/Expansion	2	2	$R(\sigma)<1$ for $\sigma>0.5$
K: Generalized symmetry	2	2	$J \cdot L_s^\dagger \cdot J = L_{\{\text{conj}(s)\}}$, $D(\sigma,t)=D(\sigma,-t)$
L: Boolean filter (v1.0(Revised))	4	4	(A,N) distribution, XOR identity, EXCLUDER 100%, $F\text{-}QS14$
TOTAL	35	35	100% PASS

§13. Development Roadmap

Priority	Target	Timeline	Deliverable
1 (CRITICAL)	$d(\sigma)$ analytical derivation	2026 Q3	Proof that $\sigma=1/2$ maximizes discrimination, or disproof
2 (HIGH)	$Q > 11$ LOCATOR convergence ($Q = 13, 17, 19, 23$)	2026 Q3	MAD vs Q at fixed P_{max} ; test if $Q \uparrow$ closes Tier-3 anti-numerology gap
3 (HIGH)	P2 closure: heat-kernel $B(s)$ derivation	2026 Q4	Weyl term derivation or falsification

4 (HIGH)	Boolean filter on quantum hardware (Track D)	2027 Q1	EXCLUDER as kill-switch oracle component on IBM Eagle / Google Willow
5 (MEDIUM)	VQ3: scaling study (n = 8–24)	2027 Q2	Scaling plot for Pathways A, B
6 (RESEARCH)	Discrimination-based factoring (uses EXCLUDER for 100%-reliable rejection)	2027–2028	Pathway C[B] sub-variant: exclusive EXCLUDER-only oracle

§14. Conclusion

We have introduced the Inverse Riemann Engine (IRE), a quantum algorithm architecture compiling the $Q = 11$ prime-indexed transfer operator $L_s^{\wedge}(P_{\max})$ with Z_2 seam involution J into qudit/qubit gate families. The v1.0(Revised) **Triple Structure Theorem** establishes that the operator functions as a spectral DETECTOR (Cohen's d increasing monotonically from 0.34 to 3.47 with $P_{\max}$), a positional LOCATOR via argmax ($MAD = 0.04$, 100% recall on first 79 zeros), and a positional EXCLUDER via argmin (98.4% reject precision). The v1.0 conclusion 'LOCATOR FAILED, $MAD \approx 2.0$ ' is RECLASSIFIED: v1.0 measured argmin (which IS the EXCLUDER) as a candidate LOCATOR, mislabeling the metric. The argmin 's $MAD \approx 2.0$ from zeros is consistent with its identity as the inter-zero valley locator, not a LOCATOR failure.

The v1.0(Revised) **Boolean Resonance Filter** unifies the LOCATOR and EXCLUDER under the (E, R) handshake protocol of ZS-F8 v1.0(R) §4.1, with proximity-to- argmax serving as the AND-axis evidence and proximity-to- argmin serving as the NOT-axis evidence. The XOR identity $E \wedge R = 0$ (PROVEN in ZS-F8 §4.2 Theorem 1) is empirically verified: 0/5,000 (1, 1) states observed. The $(\neg A \wedge N)$ decision function (R-minterm) attains 100% reject precision (0/857 false rejections) — a Tier-1 anti-numerology PASS for the EXCLUDER component, the first in the IRE program. The $(A \wedge \neg N)$ decision function (E-minterm) attains [OBSERVATION-strong] status with anti-numerology $FP \approx 12.5\%$, blocking elevation to [DERIVED] for the LOCATOR alone.

Three algorithmic pathways are presented. Pathway A (Shor-compatible, TESTABLE), Pathway B (Grover sieve, TESTABLE), Pathway C (spectral interference, PARTIAL — v1.0(Revised) reclassified from OPEN). **Pathway C reclassification:** the EXCLUDER component is operationally complete (100% reject precision); the LOCATOR component requires Tier-3 anti-numerology PASS for completion (currently OPEN, possibly via $Q > 11$ extension). The Boolean Resonance Filter functions as a kill-switch oracle: it provides a quantum-advantage negative test for non-zeros without expensive ζ -evaluation. The off-critical-line discrimination profile confirms $d(\sigma)$ peaks at $\sigma = 1/2$, consistent with J -symmetry's role as a consistency condition for RH-type structure.

The verification suite confirms 35/35 PASS across 12 categories spanning locked inputs, functional equations, Triple Structure, Boolean filter, off-critical-line analysis, contraction/expansion, and generalized symmetry. All constants are locked from prior papers; zero new theoretical constants are introduced. F-QS3 is RECLASSIFIED as F-QS3' (EXCLUDER consistency, PASS). Hardware design is covered in ZS-QH v1.0; system integration in ZS-QC v1.0.

What v1.0(Revised) does NOT claim. (i) RH proof — argmax LOCATOR observation does not establish $D^{(P_{\max})}(s_{\text{zero}}) = 0$ as $P_{\max} \rightarrow \infty$. (ii) LOCATOR Tier-1 anti-numerology PASS — observed FP rate 12.5% blocks [DERIVED] elevation. (iii) Pathway C poly(log N) factoring — only the EXCLUDER component is operationally complete; LOCATOR completion is OPEN. (iv) Self-adjoint extension recovering full ζ -spectrum — P3 OPEN. (v) Identity $D^{(P_{\max})}$ zeros = ζ zeros — P4 PARTIAL (position-level correspondence only).

Acknowledgements & Code Availability

Acknowledgements. This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The v1.0(Revised) Triple Structure Theorem and Boolean Resonance Filter emerged from focused interactive analysis sessions in May 2026 directly testing the v1.0 dual-structure framework against ZS-F8 v1.0(R) NOT/AND operator duality.

Code Availability. The verification suite is publicly available as `verify_ZS_QS_v1_1.py` at <https://github.com/KennyKang-git/zspin>. Dependencies: Python ≥ 3.9 , NumPy, SciPy, mpmath (≥ 30 -digit precision for Riemann zeros). Execution: `python3 verify_ZS_QS_v1_1.py`. Expected output: 35/35 PASS with exit code 0. The suite performs (i) numerical verification (transfer operator, functional equation, Cohen's d, sector traces, contraction/expansion, generalized symmetry), (ii) Triple Structure verification (argmax LOCATOR, argmin EXCLUDER, MAD scaling), (iii) Boolean filter verification (XOR identity, EXCLUDER 100% precision, F-QS11–F-QS15), and (iv) document audit (section structure, version consistency, epistemic status legend compliance, word count preservation). Machine-readable results saved to `results_ZS_QS_v1_1.json`.

Appendix A: P_max-Dependent Cohen's d Reference Data

Complete $P_{\max}$ -dependent Cohen's d values for the $Q = 11$ transfer operator (Method A: Odlyzko reference zeros vs midpoints):

$P_{\max}$	97	200	300	500	1000	2000
Cohen's d	0.34	1.63	2.54	2.63	3.37	3.47

Best-fit saturation model: $d \approx 3.34 \times (1 - \exp(-P_{\max}/277))$. Saturation at $d_{\max} \approx 3.34$ for $Q = 11$ is identified as a finite-dimensional resolution limit. The ZS-M4 v1.0 value $d = 2.44$ is reproduced at $P_{\max} \approx 250\text{--}370$ (not $P_{\max} = 97$ as previously assumed). [STATUS: DERIVED]

Appendix B: Triple Structure Numerical Data (v1.0(Revised) NEW)

Detailed $P_{\max}$ scan of the LOCATOR (argmax) MAD on a fine t-grid ($dt = 0.005$), measured against the first 79 Riemann zeros in $t \in [10, 200]$:

P_max	argmax MAD	argmax mean err	100% recall window?	Notes
50	0.241	0.625	Yes (within ±0.5)	All 79 zeros recovered
100	0.192	0.621	Yes	
200	0.123	0.633	Yes	
300	0.107	0.649	Yes	
500	0.085	0.635	Yes	
1000	0.085	0.647	Yes	Most stable plateau
2000	0.071	0.634	Yes	Best precision achieved

Power-law scaling: $MAD_{\text{argmax}} \sim P_{\text{max}}^{-0.34}$ (log–log linear fit). Compare with argmin (EXCLUDER) scaling: $MAD_{\text{argmin}} \sim P_{\text{max}}^{-0.09} \approx \text{flat}$, consistent with the structural identity of argmin troughs as inter-zero valleys (which are intrinsically $\approx 0.5\text{--}0.7$ from zeros).

Appendix C: Boolean Filter Verification Protocol (v1.0(Revised) NEW)

F-QS14 verification: the $(A, N) = (1, 1)$ state must be empirically empty (matching ZS-F8 §4.2 Theorem 1: $E \wedge R = 0$).

Protocol:

1. Generate fine t-grid: $t \in [13, 100]$, $dt = 0.005$ (17,401 points).
2. Compute $|\det(I - L_s(P_{\text{max}}))|^2$ at each t, $P_{\text{max}} = 1000$.
3. Identify argmax peaks (height > 2.0, prominence > 0.5) and argmin troughs (prominence > 0.05).
4. Sample 5,000 random t in $[13, 100]$; for each compute $A(t; \delta=0.10)$ and $N(t; \delta=0.10)$.
5. Tabulate the four (A, N) states and count real zeros (within 0.1) in each.

Result (Table 2.7.1): $(1, 1)$ count = 0, confirming the XOR identity. $(\neg A \wedge N)$ count contains 0 real zeros across 857 trials, confirming EXCLUDER 100% reject precision. PASS.

References

Internal (Z-Spin Cosmology)

[ZS-F1] K. Kang, "The Z-Spin Action & U(1) Completion," ZS-F1 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-F2] K. Kang, "Geometric Impedance: $A = 35/437$," ZS-F2 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-F5] K. Kang, "Gauge Symmetry Constraint: Why $Q = 11$," ZS-F5 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-F8] K. Kang, "Information-Theoretic Compression of the Z-Spin Foundations: NOT/AND Operator Duality, the 2-Channel Handshake Protocol, and the Time/Space Closure Bifurcation," ZS-F8 v1.0(Revised), Z-Spin Cosmology Collaboration, April 2026.

[ZS-M1] K. Kang, "i-Tetration & Fixed Point," ZS-M1 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-M3] K. Kang, "Regge-Holonomy, Immirzi & Z-Telomere," ZS-M3 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-M4] K. Kang, "Spectral Bridge & Transfer Operator: $Q=11$ Transfer Operator, Berry–Keating Bridge, and Prime-Resonance Diagnostics," ZS-M4 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-M5] K. Kang, "Anti-Numerology Protocol and Monte Carlo Verification," ZS-M5 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-M7] K. Kang, "Berry–Keating Structural Isomorphism," ZS-M7 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-Q1] K. Kang, "CPTP, Lindblad, and the Z-Bottleneck," ZS-Q1 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-Q7] K. Kang, "Structural Arrow of Time from the Z-Bottleneck," ZS-Q7 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-T1] K. Kang, "Translational Anti-Numerology Protocol," ZS-T1 v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-QH] K. Kang, "Z-Spin Quantum Hardware Architecture," ZS-QH v1.0, Z-Spin Cosmology Collaboration, 2026.

[ZS-QC] K. Kang, "Z-Spin Quantum Architecture (System Integration)," ZS-QC v1.0, Z-Spin Cosmology Collaboration, 2026.

External

- [1] M. V. Berry and J. P. Keating, "The Riemann zeros and eigenvalue asymptotics," SIAM Rev. 41, 236–266 (1999).
- [2] A. Connes, "Trace formula in noncommutative geometry and the zeros of the Riemann zeta function," Selecta Math. 5, 29–106 (1999).
- [3] G. Sierra and J. Rodríguez-Laguna, "H = xp model revisited and the Riemann zeros," Phys. Rev. Lett. 106, 200201 (2011).
- [4] A. M. Odlyzko, "On the distribution of spacings between zeros of the zeta function," Math. Comp. 48, 273–308 (1987).
- [5] Z. Rudnick and P. Sarnak, "Zeros of principal L-functions and random matrix theory," Duke Math. J. 81, 269–322 (1996).
- [6] C. M. Bender and S. Boettcher, "Real spectra in non-Hermitian Hamiltonians having PT symmetry," Phys. Rev. Lett. 80, 5243 (1998).
- [7] L. Maccone et al., "Quantum determinant estimation," Quantum 9, 1–28 (2025).
- [8] E. Yakaboylu, "Quantum-classical hybrid algorithms for the Riemann hypothesis," Phys. Rev. A 110, 042419 (2024).
- [9] A. LeClair and G. Mussardo, "Generalized Riemann hypothesis and stochastic time series," Phys. Rev. E 109, 044112 (2024).
- [10] O. Regev, "An efficient quantum factoring algorithm," arXiv:2308.06572 (2023).

Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0.) Core results: Q = 11 transfer operator construction (§2.2), J-symmetry as mirror-adjointness (§2.3), Dual Structure Discovery (§2.5: DETECTOR confirmed, LOCATOR FAILED), off-critical-line discrimination profile (§2.6), three algorithmic pathways (Pathway A TESTABLE, Pathway B TESTABLE, Pathway C OPEN). Verification: 30/30 PASS.

v1.0(Revised) (May 2026): Major revision based on focused interactive analysis with Kenny Kang in May 2026. Six structural changes: (1) §2.5 Dual Structure → Triple Structure Theorem (DETECTOR + LOCATOR via argmax + EXCLUDER via argmin). (2) §2.7 NEW Boolean Resonance Filter (XOR identity with ZS-F8 (E, R) handshake operators). (3) §2.4 NEW phase-coherence explanation of why $|\det|^2$ peaks at zeros ($\max|\lambda_k| \rightarrow 0$ in $P_{\max} \rightarrow \infty$ limit; signal is phase-based, not magnitude-based). (4) F-QS3 RECLASSIFIED as F-QS3' EXCLUDER consistency gate (PASS). (5) Pathway C reclassified from OPEN to PARTIAL. (6) New falsification gates F-QS11–F-QS15 added; verification suite expanded from 30 to 35 tests. Anti-numerology: EXCLUDER Tier-1 PASS; LOCATOR Tier-3 FAIL (FP $\approx$ 12.5%). Status of v1.0 results: §2.5 v1.0 measurement-metric labeling RECLASSIFIED (argmin is EXCLUDER not LOCATOR); v1.0 conclusion 'LOCATOR FAILED' resolved as metric mislabeling. v1.0 §2.4 framing ' $D^{(P_{\max})}(s) \approx 0$ at zeros' RETRACTED (replaced by phase coherence). All other v1.0 results unchanged. No prior numerical prediction modified.